Predictive Model for Hypertension Risk among Nigerian Adults Using Lifestyle Data

A Machine Learning Decision-Support Web Application

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1. Project Overview

This project is a web-based decision-support system that predicts the risk of hypertension among Nigerian adults using lifestyle and demographic data. Users register on the platform, complete a lifestyle assessment questionnaire (covering age, BMI, physical activity, smoking, alcohol, diet, stress, family history, salt intake, sleep hours, and caffeine intake), and receive an instant risk classification of Low, Moderate, or High along with a probability score and personalized health recommendations.

An administrator panel provides oversight of the system: user management, viewing all predictions made across the platform, interactive statistics charts, and a full system activity audit log.

2. Objectives

- Develop a machine learning model capable of classifying hypertension risk into Low, Moderate, and High categories.
- Build a user-friendly web application where individuals can assess their hypertension risk using lifestyle information.
- Provide personalized, actionable health recommendations based on each user's risk profile.
- Maintain prediction history so users can track their risk over time.
- Provide an administrative dashboard for user management, prediction monitoring, and system logging.

3. Technologies Used

3.1 Web Application (Frontend + Backend)

Technology	Purpose
Next.js 16 (App Router)	Full-stack framework - pages, API routes, middleware
React 19 + TypeScript	Component UI, type-safe development
Bootstrap 5.3	Responsive styling, cards, tables, forms
Recharts	Admin dashboard charts (pie, bar)
Zustand	Client-side auth state management
Zod	Input validation schemas (frontend + API)
jose	JWT signing and verification
bcryptjs	Password hashing

3.2 Machine Learning Service

Technology	Purpose
Python 3.14	ML service runtime
FastAPI	REST API exposing /predict and /health endpoints
scikit-learn	Model training (SVM selected), preprocessing pipeline
pandas + NumPy	Data handling and feature processing
joblib	Model serialization (.pkl artifact)

Pydantic	Request/response validation
pytest	API endpoint testing (6 tests)
Uvicorn	ASGI server

3.3 Database & Infrastructure

Technology	Purpose
PostgreSQL (Neon)	Cloud-hosted serverless database
Prisma 5.22	ORM - schema, migrations, queries
Vercel	Frontend + API hosting
Render	Python ML service hosting
Git/GitHub	Version control

4. System Architecture

The system follows a three-tier architecture: the Next.js web application serves the frontend and API layer, a separate Python FastAPI microservice hosts the machine learning model, and a PostgreSQL (Neon) database stores all persistent data.

[Browser] --> [Next.js App (Vercel)] --> [FastAPI ML Service (Render)]
|
v
[PostgreSQL DB (Neon)]

4.1 Prediction Flow

- User submits the lifestyle assessment form (11 inputs).
- Next.js API route authenticates the user (JWT cookie) and validates input with Zod.
- The input is forwarded to the FastAPI ML service, which returns a probability and risk level.
- The lifestyle data and prediction result are saved to PostgreSQL, and the action is logged.
- The UI displays the risk level, probability bar, input summary, and tailored recommendations.

4.2 Authentication & Access Control

A single login form serves both regular users and administrators. A 'role' field on the User table distinguishes them. JWT tokens carry the role, and three layers of protection secure the app: middleware (route-level), layout components (client-side), and API routes (session verification). Admins redirect to /admin; regular users to /dashboard.

5. Machine Learning Model

5.1 Input Features (11)

Age, BMI, Physical Activity, Smoking Status, Alcohol Consumption, Diet Quality, Stress Level, Family History, Salt Intake, Sleep Hours, Caffeine Intake.

5.2 Training

A synthetic dataset of 5,000 Nigerian-adult lifestyle records was generated using rule-based weighted risk scoring.

Five candidate algorithms were trained and compared: Logistic Regression, Decision Tree, Random Forest, Support Vector Machine (SVM), and a Multi-layer Perceptron (Neural Network).

5.3 Selected Model - SVM

Metric	Score
Accuracy	0.782
Precision	0.774
Recall	0.682
F1 Score	0.725
ROC-AUC	0.862

The trained model together with its preprocessing pipeline (OrdinalEncoder + StandardScaler) is serialized to a single artifact (hypertension_model.pkl) deployed with the ML service.

6. Application Features

6.1 User Features

- Registration and login with validation and password visibility toggle.
- Hypertension risk assessment form with 11 inputs, inline validation, and loading states.
- Result display: color-coded risk card, animated probability bar, input summary.
- Personalized health recommendations driven by risk level and individual inputs.
- Prediction history with filters (All/Low/Moderate/High) and a detail modal.
- Profile management - edit name/gender, change password with verification.

6.2 Admin Features

- Dashboard with stat cards and charts: risk distribution pie, gender split pie, 7-day predictions bar chart, average risk score.
- User management: searchable table, delete users with confirmation modal.
- View all predictions across all users with risk filter and user search.
- System logs viewer with status filters (Success/Failed/Error).

6.3 UI/UX Polish

- Responsive layout for mobile, tablet, and desktop (horizontal-scroll sidebar on mobile).
- Toast notifications, skeleton loaders, animated buttons and modals.
- Role-aware navbar showing a User/Admin badge and profile dropdown.

7. Database Design

Four tables implemented via Prisma on PostgreSQL:

Table	Purpose	Key Fields
User	Accounts (users + admins via role field)	email, passwordHash, role
LifestyleData	Submitted lifestyle inputs	11 lifestyle fields
PredictionResult	Prediction outcomes	riskProbability, riskLevel
SystemLog	Activity audit trail	activity, status, logTime

8. Deployment

Component	Platform	URL
Web App (Next.js)	Vercel	predictive-model-chi.vercel.app
ML Service (FastAPI)	Render	hypertension-ml-service.onrender.com
Database	Neon PostgreSQL	serverless (managed)

The Next.js app is built with 'prisma generate && next build'. The ML service is deployed via a Render Blueprint (render.yaml) which runs 'uvicorn app.main:app' against the committed model artifact.

9. Scope

Included: lifestyle-based hypertension risk prediction for Nigerian adults, full web UI, admin panel, model training and evaluation, personalized recommendations, prediction history, and user management.

Excluded: clinical diagnosis, real-time monitoring, hospital EMR integration, wearable device sync, and lab/ECG/genetic inputs. The model is trained on synthetic data and is intended as a decision-support demonstration, not a medical device.

10. Demo Credentials

Role	Email	Password
User	john@example.com	password123
Admin	admin@healthpredict.com	admin123

This document summarizes the implementation of a BSc Computer Science final-year project. The complete source code, dataset generator, and deployment configurations are available in the project repository.